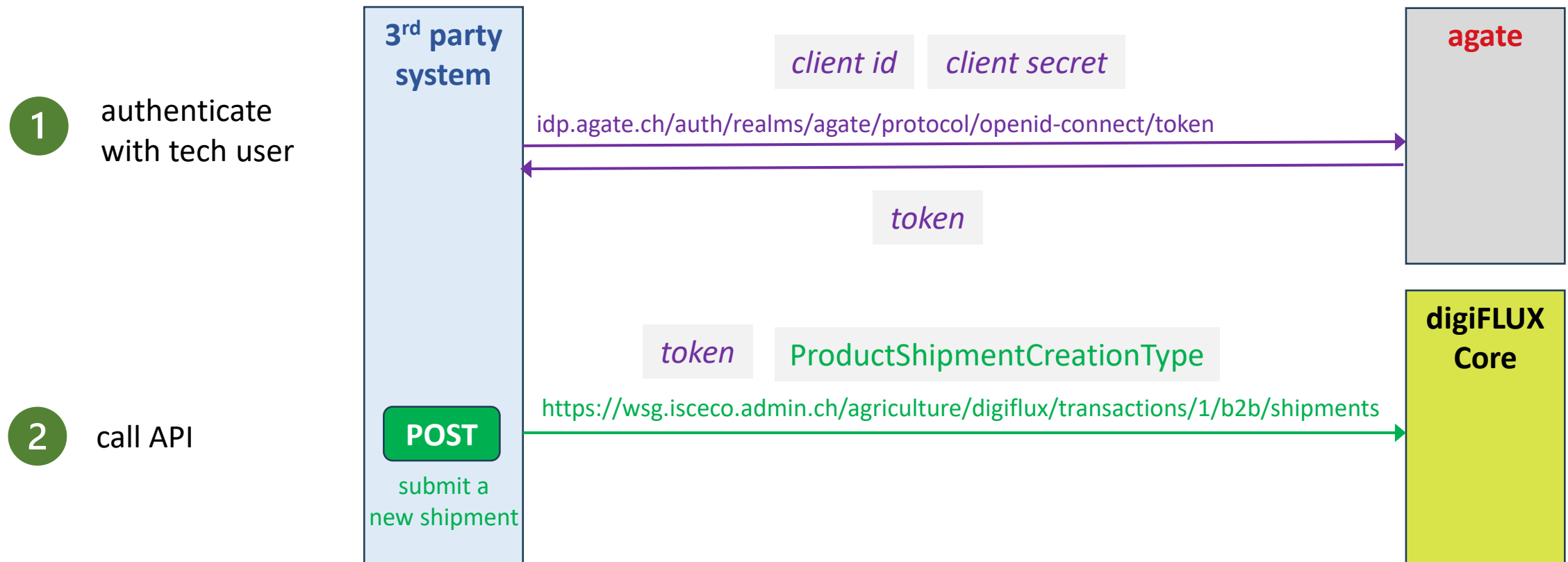
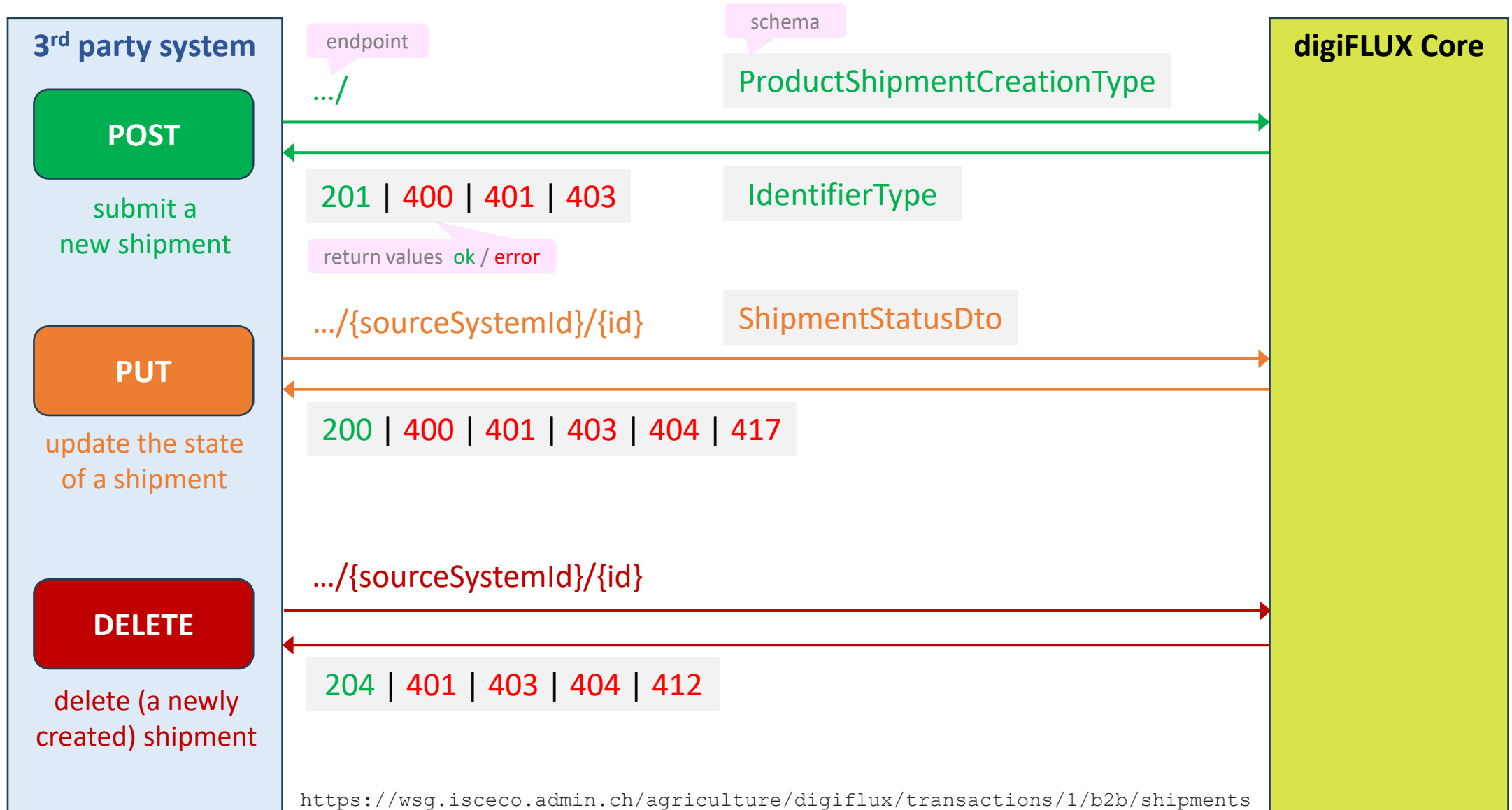


Using the digiFLUX API

Accessing the system



create / update / delete shipments



For endpoint and schema details use e. g. Swagger...

Product Shipment Update Resource ^

POST /b2b/shipments
Add and store a new 'shipment' transaction.

Creates a new shipment entry. The id can be omitted, as it is ignored

Parameters

No parameters

Request body application/json

Example Value | Schema

```
{
  "shippedFromStock": {
    "id": "string",
    "sourceSystemId": "string"
  },
  "shippedToStock": {
    "id": "string",
    "sourceSystemId": "string"
  },
  "product": {
    "id": "string"
  }
}
```

SCHEMA InventoryAddress

SCHEMA Address

SCHEMA ShippedStock

SCHEMA LocalDate

SCHEMA MeasurementType

SCHEMA ApplicationTarget

SCHEMA ProductShipmentCreationType

SCHEMA ProductShipmentType

SCHEMA ShipmentStatusDto

SCHEMA SortOrder

<https://app.swaggerhub.com/apis/blwadminch/digiflux-trans-api/0.1>

<https://app.swaggerhub.com/apis/blwadminch/digiflux-stocks-api/1.0.0>

... and see the documentation on GitHub

```
ProductShipmentCreationType {
  description: Entity representing the creation details
    of a product shipment. It includes
    information such as the external ID,
    product details, stock information,
    shipment date, status, comments,
    rejection reasons, backup address, and
    flags indicating if the delivery is a
    return or cancel.

  shippedFromStock IdentifierType > {...}
  shippedToStock IdentifierType > {...}
  product* > {...}
  productFamily > [...]
  quantity* > {...}
  shipmentDate* > [...]
  alternateId > {...}
  deliveryNumber > [...]
  comment > [...]
  shippedFromAddress > {...}
  shippedToAddress > {...}
  proxyAddress > {...}
  specialDeliveryType > [...]
  affectedDelivery > {...}
}
```

Swagger extract

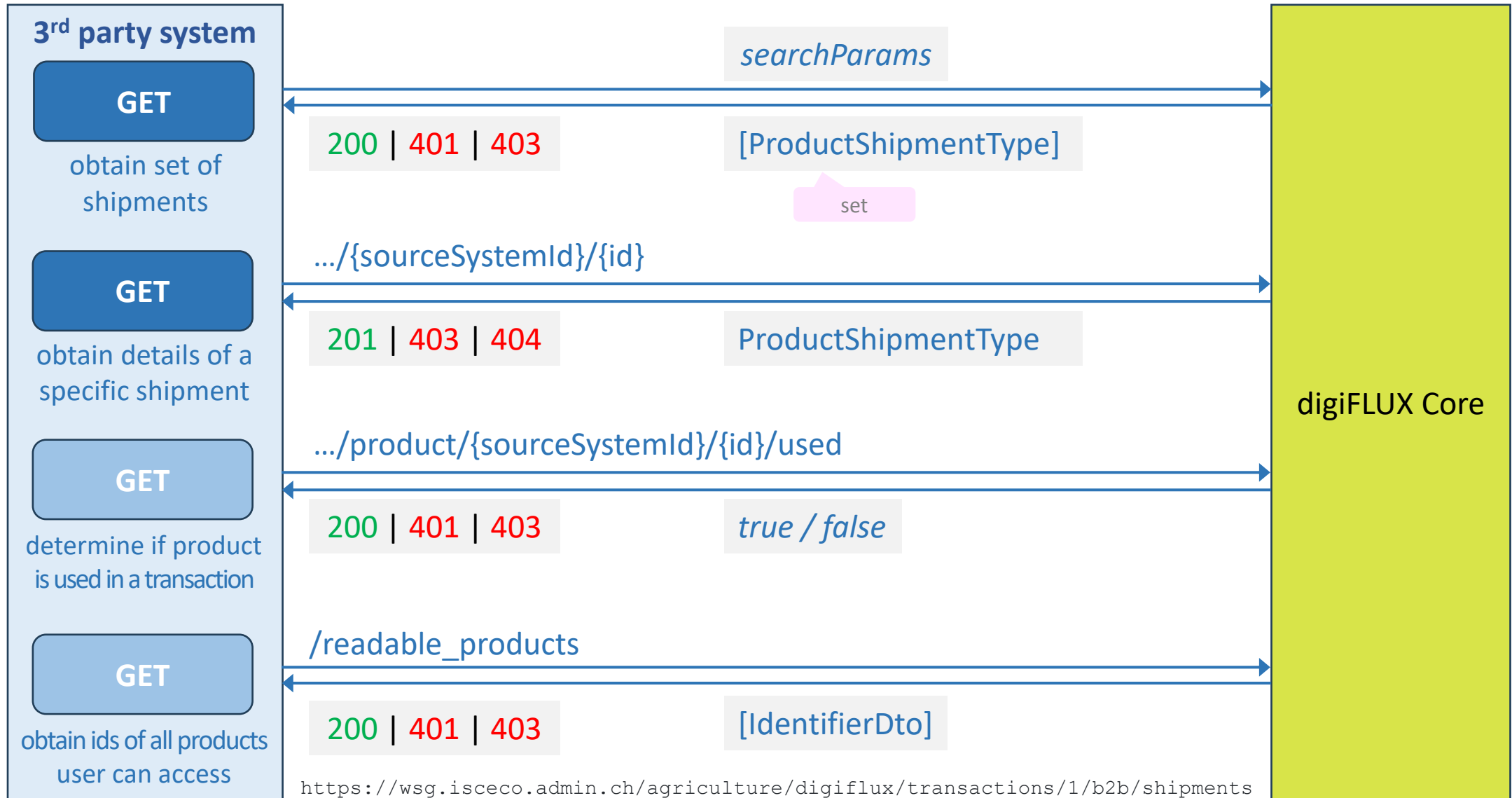
description of ProductShipmentCreationType structure

attribute	meaning	type	cardinality	unique
shippedFromStock	default form to specify the stock of the sender (typically identified by their BUR)	structure	0-1	-
- sourceSystemId	BUR / DFLID	text	1	-
- id	the sender's BUR	text	1	-
shippedToStock	default form to specify the stock of the recipient (typically identified by their DFLID)	structure	0-1	-
- sourceSystemId	BUR / DFLID	text	1	-
- id	the sender's BUR	text	1	-
product	the product being delivered	structure	1	-
- sourceSystemId	system that generated the identifier(INFOFITOZ / RPC , etc. see below)	text	1	-

GitHub extract

→ <https://github.com/blw-ofag-ufag/digiFLUX/wiki/digiFLUX-Core>

retrieve shipments



obtain DFLID of a stock



<https://wsg.isceco.admin.ch/agriculture/digiflux/stocks/1/b2b/localUnits/?name=BLW>

Return values for example `?name=BLW`

```
[{"dfid":"D02384801","name":"BLW Testbetrieb Biogas",
"nameBusiness":"Vergärungs- und Kompostierungsanlage sowie
Nährstoffpool","swissZipcode":"8001","town":"Zürich",
"street":"","houseNumber":"","uid":"CHE114587210"},
{"dfid":"D02384744","name":"BLW Testbetrieb Landwirtschaft",
"nameBusiness":"Vergärungs- und Kompostierungsanlage sowie
Nährstoffpool","swissZipcode":"8001","town":"Zürich",
"street":"","houseNumber":"","uid":"CHE114587210"}]
```

```
JSON
0
  dfid : "D02384801"
  name : "BLW Testbetrieb Biogas"
  nameBusiness : "Vergärungs- und Kompostierungsanlage sowie Nährstoffpool"
  swissZipcode : "8001"
  town : "Zürich"
  street : ""
  houseNumber : ""
  uid : "CHE114587210"
1
  dfid : "D02384744"
  name : "BLW Testbetrieb Landwirtschaft"
  nameBusiness : "Vergärungs- und Kompostierungsanlage sowie Nährstoffpool"
  swissZipcode : "8001"
  town : "Zürich"
  street : ""
  houseNumber : ""
  uid : "CHE114587210"
```

Script showcasing shipments services

https://github.com/blw-ofag-ufag/digiFLUX/blob/main/resources/digiFLUX_API_testscript_1_0.sh

The script simulates an end-to-end shipment process involving a **sender** and a **recipient**. The sender creates the shipment entry, the recipient accepts it and, finally, the sender checks the state of the shipment.

step by step processing

- 1 authenticate with tech user 1 as sender
- 2 authenticate with tech user 2 as recipient
- 3 submit shipment as sender using POST
- 4 wait 120 secs (until shipment has automatically changed from state CREATED to SENT (and can thus be accepted by recipient
- 5 accept delivery as recipient using PUT
- 6 check state of delivery as sender using GET

usage variants

- dry-run mock response (no https calls)
- verbose print full API responses
- help show help message

sample output

```
$ sh digiFLUX_API_testscript_1_0.sh
[INFO] Fetching sender token...
[INFO] Fetching recipient token...
-----
[INFO] Sending delivery POST...
[OK] Delivery created.
-----
[INFO] Checking delivery state...
[INFO] Initial state: CREATED
-----
[INFO] Waiting 120s for background update...
[INFO] Updated status: SENT
-----
[INFO] Sending PUT to accept delivery...
[OK] Delivery accepted by recipient.
-----
[INFO] Confirming final delivery status...
[INFO] Final delivery state: ACCEPTED
[OK] Delivery successfully marked as ACCEPTED.
```

Script showcasing dflid search

https://github.com/blw-ofag-ufag/digiFLUX/blob/main/resources/digiFLUX_API_stocks_testscript_1_0.sh

The script uses the localUnits service to obtain the dflids that match a selected search criterion, specifically «?name=BLW». The dflid attribute is primarily used to specify the recipient of a shipment.

step by step processing

- 1 authenticate with tech user
- 2 Call GET localUnit with search param «?name=BLW»

usage variants

- | | |
|------------|--------------------------------|
| -- dry-run | mock response (no https calls) |
| -- verbose | print full API responses |
| -- help | show help message |

sample output

```
[INFO] Fetching sender token...
-----
[INFO] get local units...
[{"dflid":"D02384801","name":"BLW Testbetrieb Biogas","nameBusiness":"Vergärungs- und Kompostierungsanlage sowie Nährstoffpool","swissZipcode":"8001","town":"Zürich","street":"","houseNumber":"","uid":"CHE114587210"}, {"dflid":"D02384744","name":"BLW Testbetrieb Landwirtschaft","nameBusiness":"Vergärungs- und Kompostierungsanlage sowie Nährstoffpool","swissZipcode":"8001","town":"Zürich","street":"","houseNumber":"","uid":"CHE114587210"}]
```

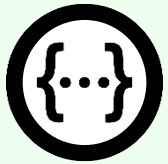

Resources



openapi files for shipments and stocks services

<https://rf-vp.agate.ch/digiflux/core/transactions/q/publicapi>

<https://rf-vp.agate.ch/digiflux/core/stocks/q/publicapi>



Swagger documentation of shipments and stocks services

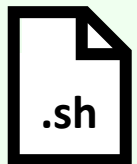
<https://app.swaggerhub.com/apis/blwadminch/digiflux-trans-api/0.1>

<https://app.swaggerhub.com/apis/blwadminch/digiflux-stocks-api/1.0.0>



GitHub documentation

github.com/blw-ofag-ufag/digiFLUX/wiki/digiFLUX-Core



sample shell script to submit, accept and check shipment

https://github.com/blw-ofag-ufag/digiFLUX/blob/main/resources/digiFLUX_API_testscript_1_0.sh



sample shell script to obtain the dflid of a stock

https://github.com/blw-ofag-ufag/digiFLUX/blob/main/resources/digiFLUX_API_stocks_testscript_1_0.sh